

Servas Adolph Tarimo

PhD Researcher in Future Convergence Technology / Big Data Engineering
Soonchunhyang University, South Korea

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RESEARCH PROFILE

PhD researcher working at the intersection of medical image analysis, domain adaptation, multimodal report generation, and applied LLM/RAG systems. My research focuses on building AI models that remain reliable across hospitals, imaging devices, clinical protocols, and patient populations, with long-term interest in practical healthcare tools for Tanzania and other low-resource settings.

EDUCATION

Ph.D. Student, Future Convergence Technology / Big Data Engineering 2023.09 - Present

Soonchunhyang University, South Korea

- Advisor: Prof. Woo Ji-Young.
- Lab: Advanced Data Mining Lab (ADM Lab).

M.Sc. Big Data Engineering 2021.09 - 2023.08

Soonchunhyang University, South Korea

- Dissertation: WBC YOLO-ViT: 2-Way 2-Stage White Blood Cell Detection and Classification with a Combination of YOLOv5 and Vision Transformer.
- Published paper | Thesis PDF.
- Advisor: Prof. Woo Ji-Young.
- Lab: Advanced Data Mining Lab (ADM Lab).

B.Sc. Computer Engineering & Information Technology 2017.01 - 2020.08

United African University of Tanzania

RESEARCH EXPERIENCE

Advanced Data Mining Lab, Soonchunhyang University 2023 - Present

Advisor: Prof. Woo Ji-Young

- Domain Adaptation in Medical AI - developing transfer learning and domain adaptation methods for generalizing medical imaging and healthcare models across clinical sites, imaging protocols, and patient populations.
- Medical Report Generation & RAG for Clinical Decision Support - building LLM-based systems that produce structured clinical reports and physician-style patient summaries grounded in verified medical literature via Retrieval-Augmented Generation.
- Current directions include domain-robust medical models, multimodal report generation for blood smear analysis, and practical AI systems that remain trustworthy beyond a single dataset.

Advanced Data Mining Lab, Soonchunhyang University 2021 - 2023

Advisor: Prof. Woo Ji-Young

- White Blood Cell Classification and Segmentation - conducted research on automated blood-cell detection, counting, and classification using YOLO, Vision Transformers, and classical machine-learning pipelines.
- Generative Models for Medical Data - developed data augmentation approaches for limited blood-cell image datasets, including GAN-based image generation for class-imbalance mitigation.

WORK EXPERIENCE

Research Assistant

2021.09 - Present

Advanced Data Mining Lab (ADM Lab), Soonchunhyang University, South Korea

- Developing medical AI systems for blood-cell image analysis, domain adaptation across clinical sites, multimodal report generation, and RAG-based clinical decision-support workflows with Prof. Woo Ji-Young.

SELECTED PROJECTS

Domain Adaptation in Medical AI 2026 - Present	Transfer learning and domain adaptation methods for robust medical AI across hospitals, imaging devices, and patient populations.
Multimodal Report Generation for Blood Smear Analysis 2026 - Present	AI systems that read microscopy blood-smear images and generate structured clinical reports using attention mechanisms and language models.
Matokeo Yangu 2026, completed	Bilingual platform helping Tanzanian students check exam results and receive AI-guided university and career-pathway advice.
WBC Detection & Classification 2021.09 - 2024, completed	Automated white blood cell detection, counting, and classification using YOLO and Vision Transformers for resource-constrained clinical laboratories.

PUBLICATIONS

Yunjung Hong, Servas Adolph Tarimo, and Jiyoung Woo. **Pancreas Segmentation Using a Two-Stage Pipeline of Faster R-CNN and TransUNet**. *Applied Sciences*, 16(12):5764, 2026. doi:10.3390/app16125764.

Servas Adolph Tarimo, Mi-Ae Jang, Emmanuel Edward Ngasa, Hee Bong Shin, HyoJin Shin, and Jiyoung Woo. **WBC YOLO-ViT: 2-Way 2-Stage White Blood Cell Detection and Classification with a Combination of YOLOv5 and Vision Transformer**. *Computers in Biology and Medicine*, 169:107875, 2024. doi:10.1016/j.combiomed.2023.107875.

Emmanuel Edward Ngasa, Mi-Ae Jang, Servas Adolph Tarimo, Jiyoung Woo, and Hee Bong Shin. **Diffusion-based Wasserstein Generative Adversarial Network for Blood Cell Image Augmentation**. *Engineering Applications of Artificial Intelligence*, 133:108221, 2024. doi:10.1016/j.engappai.2024.108221.

Servas Adolph Tarimo. **Adapting YOLO-ViT for Differential Diagnosis of Myelodysplastic Syndromes and Normal Blood Cell**. *Proceedings of the Korea Society of Computer and Information Conference*, 2024.

Servas Adolph Tarimo and Jiyoung Woo. **White Blood Cell Detection and Classification using YOLOv5 with Hybrid ResNet50-VGG16-SVM**. *Proceedings of the 6th International Conference on ICT for Smart Health & Home*, Kota Kinabalu, Malaysia, 2022.

TECHNICAL COMPETENCIES

- Programming and frameworks: Python, PyTorch, FastAPI, React PWA, OpenCV.
- Medical AI: blood-cell image analysis, pancreas segmentation, detection, classification, report generation.
- Methods: domain adaptation, transfer learning, YOLO, Vision Transformers, GANs, RAG, LLM-based decision-support workflows.
- Data systems: Supabase/PostgreSQL, hybrid search, structured clinical documentation, EHR data analysis.

RESEARCH INTERESTS

Domain adaptation in medical AI; cross-site model generalization; blood-smear image analysis; multimodal clinical report generation; retrieval-augmented generation for decision support; practical AI systems for low-resource healthcare and education contexts.

SELECTED LINKS

Portfolio | GitHub | Google Scholar | ORCID | LinkedIn | YouTube

Last updated: August 2026